MPTA-Repair: Simultaneous Multi-Property Clock Bound Repair for Networks of Timed Automata via Iterative Space Optimization

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***Abstract*—Networks of timed automata (NTA) are widely used to model and verify real-time industrial systems, which are typically required to satisfy multiple properties simultaneously. When verification fails, repairing an NTA is challenging because a valid repair must modify erroneous clock constraints, satisfy all properties, and preserve the original functional behavior. We present MPTA-Repair, an automated repair framework for multi-property NTA. It uses an iterative, counterexample-guided MaxSMT approach to generate minimally modified candidates, exploiting relations among properties, counterexamples, and repair variables to optimize search. Each candidate is validated by full-property regression verification and an admissibility check based on functional equivalence. Compared to the TARTAR baseline, MPTA-Repair achieves higher effectiveness on faulty models generated via our proposed mutation strategy. Specifically, it improves the success rate from 72% to 93% on benchmarks derived from UPPAAL and the XtaBenchmarkSuite, and from 20% to 76% on an industrial dataset constructed in this work.**

***Index Terms*—networks of timed automata, automated model repair, MaxSMT, multi-property verification**

I. Introduction

Real-time systems are widely used in domains such as railway control, communication protocols, and aerospace scheduling [1]–[3]. Their correctness depends not only on whether they exhibit the intended discrete behavior, but also on whether relevant actions occur within the required time bounds. Timed automata (TAs) provide a formalism for modeling these two aspects by extending finite-state automata with real-valued clocks and timing constraints that govern both the occurrence of transitions and the time spent in locations [4], [5]. Systems composed of multiple interacting components are commonly represented as networks of timed automata (NTA). In practice, NTA are typically verified against sets of properties that collectively specify the required system behavior, including safety and liveness requirements [6], [7].

Among the properties checked for an NTA, this work focuses on timed safety properties. This scope is motivated by the diagnostic basis of clock-bound repair: violations of such properties can be witnessed by finite timed executions, which provide concrete evidence for subsequent repair analysis [7], [8]. Timed safety properties specify bounded timing requirements, such as deadlines, limits on waiting times, and valid execution durations. Guaranteed timed safety presents a significant challenge in complex NTAs with numerous interacting components. In these models, multiple real-valued clocks advance simultaneously, and their constraints determine when transitions may occur and how long the system may remain in a location. As a result, an incorrect numerical clock bound may affect executions across multiple components, causing an action to occur too early, too late, or for an invalid duration, and thereby violating a timed safety property [2], [8], [9]. When such a violation is detected, the erroneous clock bounds must be repaired while preserving the intended functional behavior of the system.

When a property violation is detected, real-time model checkers such as UPPAAL, Kronos, and opaal can return a timed counterexample, commonly called a timed diagnostic trace (TDT) [10]–[12]. A TDT describes a timed sequence of transitions and delays that leads the NTA from its initial state to a state in which the property is violated. It shows how the violation occurs, but does not explain which clock bounds caused it or how those bounds should be changed. This repair problem becomes more difficult when an NTA must satisfy multiple properties simultaneously. Even if a candidate clock-bound change resolves the violation witnessed by one TDT, other violations may remain, and the same change may cause a previously satisfied property to fail or compromise the original functional behavior. Therefore, a candidate repair cannot be accepted solely because it resolves an individual violation; it must also be validated against the full property set and checked for admissibility based on functional equivalence.

Parameterization and constraint solving provide a general basis for repairing timing constraints in timed automata. A common strategy is to replace selected clock bounds with parameters and synthesize valuations under which the resulting model satisfies a specified property. Knapik and Penczek encode bounded executions of parametric timed automata as SMT formulas and synthesize parameter valuations for reachability properties [13]. TARTAR [8] encodes the timing constraints of a TDT in linear real arithmetic and uses MaxSMT to generate candidate repairs with minimal changes to clock bounds. It then performs an admissibility check based on functional equivalence to determine whether a candidate preserves the original functional behavior. Although these methods show that parameterization and constraint solving

The implementation and datasets are publicly available at https://github.com/zhougut/MPTA-Repair.

can support automated model repair, their repair or synthesis procedures are formulated for an individual property or TDT rather than a property set that must be satisfied simultaneously. As a result, they do not provide an iterative repair strategy or search optimization specifically designed for the multi-property setting.

To address this limitation, we present MPTA-Repair, an automated framework for simultaneous multi-property clock-bound repair of NTAs. The main contributions of this work are as follows. First, we formalize simultaneous multi-property clock-bound repair for NTA models, together with the validity conditions of full-property satisfaction and preservation of the original functional behavior. Second, we propose an iterative, counterexample-guided MaxSMT framework that maintains a joint property-trace constraint pool and exploits relations among properties, counterexamples, and repair variables to optimize the repair search. Each candidate repair model undergoes full-property regression verification and an admissibility check based on functional equivalence. If regression verification reveals a new violation, the corresponding TDT is added to the property-trace constraint pool, and the updated pool is used in the next repair iteration. Third, we propose a mutation strategy for constructing faulty NTA models and evaluate MPTA-Repair against the TARTAR baseline [8] on benchmark models from UPPAAL examples [14] and the XtaBenchmarkSuite [15], as well as industrial models constructed in this work, showing higher repair success rates in both settings.

The remainder of the paper is organized as follows. Section II introduces the preliminaries and problem setting. Section III presents MPTA-Repair. Section IV reports the experimental evaluation. Section V concludes the paper.

II. Preliminaries

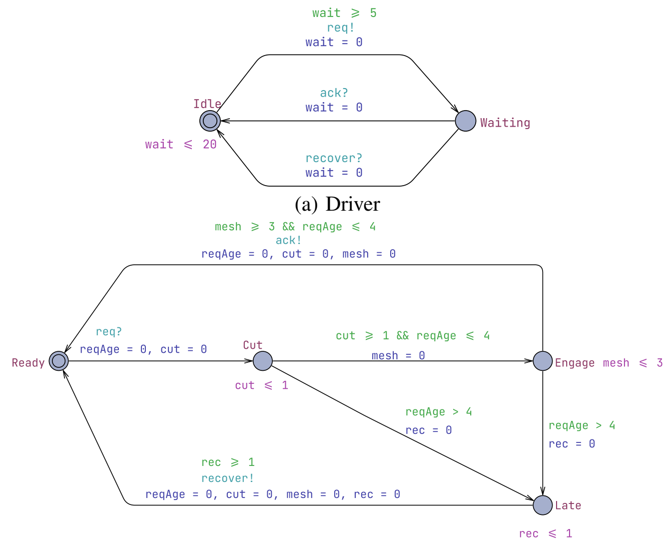
A. Networks of Timed Automata

The timed automaton model used in this paper follows the standard timed-automata semantics [5].

Definition 1. A timed automaton (TA) is a tuple

$$T = (L, \ell_0, C, \Sigma, \Theta, I) \tag{1}$$

where L is a finite set of locations, ℓ_0 is the initial location, C is a finite set of clocks, Σ is a set of action labels, Θ is a finite set of transitions, and I maps each location to a clock invariant. A clock constraint is a conjunction of atoms $x \sim b$, where $x \in C$, $\sim \in \{<, \leq, =, \geq, >\}$, and $b \in \mathbb{R}_{\geq 0}$. A transition $\theta = (\ell, g, a, R, \ell')$ moves from ℓ to ℓ' when guard g is satisfied, emits action a, and resets clocks in $R \subseteq C$. An NTA is a parallel composition $\mathcal{N} = T_1 \parallel \cdots \parallel T_k$ with the synchronization semantics of the target modeling language, such as UPPAAL [10]. Its behavior consists of action and delay transitions. Action transitions are instantaneous, require their guards to be satisfied, and reset the specified clocks; delay transitions model the passage of time in a location while respecting its invariant.



(a) Driver

(b) GearController

Fig. 1. Simplified gear-shift example.

B. Timed Diagnostic Traces (TDTs) and Multi-Property Motivation

This paper focuses on timed safety properties, which are commonly used to specify deadlines, bounded waiting, and reachable error states in real-time systems. Such properties can be checked by tools such as UPPAAL: given an NTA and a property, the verifier either confirms the property or returns a counterexample trace [7].

Definition 2. A symbolic timed trace (STT) of a model is a finite transition sequence $S = \theta_0, \ldots, \theta_{n-1}$. A realization of S is a sequence of non-negative delays $d_0, \ldots, d_n$ and states $s_0, \ldots, s_{n+1}$ such that, for every $i < n$, the model can delay from s_i by d_i and then take θ_i to s_{i+1} while respecting guards, resets, invariants, and synchronizations; the final delay d_n leads from s_n to s_{n+1}. The STT is feasible if it has at least one realization. Given a safety property φ, a feasible STT is a TDT for φ if at least one realization reaches a state that violates φ.

Example 1. Figure 1 shows the running gear-shift example. We check φ_1, `A[] (Controller.Engage imply cut >= 2)`, and φ_2, `A[] not Controller.Late`. In the faulty controller, the `Cut` invariant $cut \leq 1$ and the `Cut`$\rightarrow$`Engage` guard $cut \geq 1$ allow engagement after one time unit, violating φ_1. A single-property repair raises both cut bounds to 2. If the `Engage` invariant $mesh \leq 3$ and the `Engage`$\rightarrow$`Ready` guard $mesh \geq 3$ remain unchanged, the controller cannot return to `Ready` until $2 + 3 = 5$ time units after the request. It can therefore reach `Controller.Late` and violate φ_2. A multi-property repair must therefore adjust both cut and $mesh$ while preserving the original functional behavior.

TDTs provide concrete realizations of property violations, but they do not explain why the violations occur. In particular, a TDT does not identify which numerical bound is faulty or whether the relevant bound appears in a guard or an invariant. Following the idea of TARTAR [8], the repair procedure developed later in this paper encodes TDTs as arithmetic constraints over trace feasibility, property violation, and bound variations. Unlike single-property repair, MPTA-Repair organizes constraints from multiple violated properties and TDTs into a joint solving process to generate candidate clock-bound repairs.

Let M_{bad} be a faulty NTA model and let $\Phi = \{\varphi_1, \ldots, \varphi_m\}$ be the set of properties that must be satisfied simultaneously. MPTA-Repair generates a repaired model M_{rep} by changing repairable numerical bound occurrences in guards and invariants. A candidate repair is accepted only if

$$M_{rep} \models \bigwedge_{\varphi \in \Phi} \varphi \quad \wedge \quad \mathrm{Accept}(M_{bad}, M_{rep}). \tag{2}$$

The first condition requires the repaired model to satisfy all required properties. The second condition denotes a TARTAR-style admissibility check based on functional equivalence [8]. This check prevents repairs that satisfy properties merely by disabling required actions.

III. Approach

A. Overview

MPTA-Repair takes an NTA model M and a property set $\Phi = \{\varphi_1, \ldots, \varphi_m\}$ as input. Instead of requiring manual variable specification, the framework extracts repairable bounds from numerical clock constraints in transition guards and location invariants. The final output is either a repaired model M' accompanied by concrete clock-bound edits, or a failure report if no valid full-property repair can be synthesized within the resource budget.

The framework operates on a local-generation and global-validation principle. While each TDT drives candidate generation, validation remains global to ensure that fixing one violation does not introduce regressions elsewhere. To leverage the interdependence where multiple violations share identical timing structures, MPTA-Repair maintains a joint pool of property-trace constraints. This joint structure enables the framework to exploit relations among properties, counterexamples, and repairable bounds to optimize the search space and guide the MaxSMT solver toward minimal parameter assignments.

Operationally, each refinement round begins by verifying M against Φ to collect TDTs for any violations. If violations are detected, the framework identifies active repairable bounds and constructs a joint MaxSMT problem from the active constraint pool. The synthesized parameter assignment is instantiated into a candidate model, which undergoes global validation via full-property regression verification and an admissibility check based on functional equivalence. A rejected candidate contributes newly exposed TDTs back into the pool to further constrain subsequent rounds, whereas a candidate passing both checks is returned as the valid repair M'. Figure 2 summarizes this counterexample-guided refinement loop from verification and relation-guided MaxSMT solving to global validation.

B. Joint Trace Encoding and Optimization Objective

MPTA-Repair follows the clock-bound variation strategy used in TARTAR [8]. Let S denote the set of repairable clock-bound occurrences. For each occurrence $s \in S$, the original numerical bound b_s in a transition guard or location invariant is associated with a variation variable δ_s. This ensures that the repaired constraint retains the original clock reference and comparison operator while incorporating a shifted value:

$$b'_s = b_s + \delta_s \tag{3}$$

The repair space is therefore strictly restricted to clock-bound constants, keeping transition structures and discrete updates unaltered.

MPTA-Repair builds on the trace-encoding formulation of TARTAR and extends it to the multi-property setting [8]. Given a TDT

$$\begin{aligned} \tau = \ell_0 &\xrightarrow{d_0, e_0} \ell_1 \cdots \\ &\xrightarrow{d_{n-1}, e_{n-1}} \ell_n \end{aligned} \tag{4}$$

a trace path is encoded as a conjunction of linear arithmetic constraints:

$$\begin{aligned} \mathrm{Path}(\tau, \rho, \mathbf{z}_\tau) = {} & \mathit{Init}(\nu_0) \\ & \wedge \bigwedge_{i=0}^{n-1} \Big(d_i \geq 0 \wedge \mathit{Inv}(\ell_i, \nu_i + d_i, \rho) \\ & \wedge \mathit{Guard}(e_i, \nu_i + d_i, \rho) \\ & \wedge \nu_{i+1} = \mathit{Reset}(\nu_i + d_i, R_i) \Big) \end{aligned} \tag{5}$$

Here, ρ denotes the modified bounds, and $\mathbf{z}_\tau$ collects the delays and clock valuations. The reset equation $\nu_{i+1} = \mathit{Reset}(\nu_i + d_i, R_i)$ updates the valuation after transition e_i by resetting clocks in R_i and preserving the others. The original violation encoding combines path executability with the final property violation:

$$\mathrm{Enc}(\tau, \varphi, \rho) = \mathrm{Path}(\tau, \rho, \mathbf{z}_\tau) \wedge \mathit{Bad}_\varphi(\ell_n, \nu_n) \tag{6}$$

The predicate $\mathit{Bad}_\varphi(\ell_n, \nu_n)$ characterizes the violation state of φ at the final location. To block this violating behavior while preserving executability of the discrete path skeleton, the single-trace repair constraint uses two copies of the trace variables:

$$\begin{aligned} \Psi^{\mathrm{repair}}_{\tau,\varphi}(\rho) = {} & \exists \mathbf{z}^+_\tau .\, \mathrm{Path}(\tau, \rho, \mathbf{z}^+_\tau) \\ & \wedge \neg \exists \mathbf{z}^-_\tau .\Big(\mathrm{Path}(\tau, \rho, \mathbf{z}^-_\tau) \\ & \wedge \mathit{Bad}_\varphi(\ell^-_n, \nu^-_n) \Big) \end{aligned} \tag{7}$$

The first part requires that the original discrete path skeleton still has at least one timed realization after repair. The second part requires that no assignment of delays and clock valuations can make that same path violate φ.

We extend this single-trace formulation to a joint, multi-property incremental setting. At refinement round k, counterexamples from all currently violated properties are accumulated into a trace pool

$$\mathcal{T}_k = \{(\tau_j, \varphi_j)\}_{j=1}^{m_k} \tag{8}$$

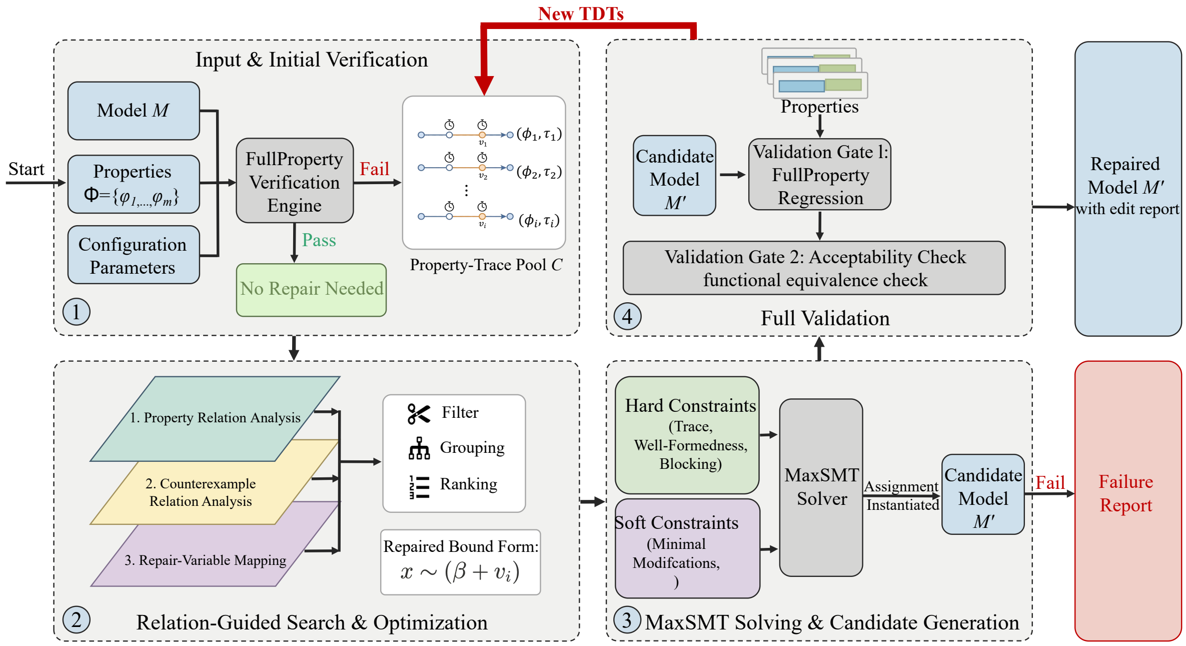


Fig. 2. Overview of the MPTA-Repair workflow.

MPTA-Repair assembles a joint MaxSMT problem with the constraint system defined as

$$\begin{aligned}\Phi_k(\rho) = &\mathit{Dom}(\rho) \wedge \mathit{Block}_k(\rho)\\ &\wedge \bigwedge_{(\tau,\varphi)\in \mathit{Select}(\mathcal{T}_k)} \Psi^{\text{repair}}_{\tau,\varphi}(\rho)\end{aligned} \tag{9}$$

Here, $\mathit{Dom}(\rho)$ enforces static interval consistency and domain limits, such as $b'_s \geq 0$, and $\mathit{Select}(\mathcal{T}_k)$ represents a reduced subset of traces obtained after relation-aware filtering. When an instantiated candidate model fails verification, the constraint system is updated incrementally. The framework expands the pool via

$$\mathcal{T}_{k+1} = \mathcal{T}_k \cup \mathit{NewTDTs}(M(\rho_k), \Phi) \tag{10}$$

and updates the search-space block to exclude the failed assignment ρ_k over the set of repairable bounds S:

$$\mathit{Block}_{k+1}(\rho) = \mathit{Block}_k(\rho) \wedge \left(\bigvee_{s\in S} \delta_s \neq \delta_s^k\right) \tag{11}$$

where δ_s^k is the valuation assigned to variation variable δ_s in round k.

The optimization objective is governed by a hierarchical set of soft constraints. MPTA-Repair uses a lexicographic objective that first favors small repairs and then incorporates risk- and coverage-based priorities:

$$\operatorname*{lex\,min}_{\rho} \begin{pmatrix} \sum_{s\in S} \mathbf{1}\{\delta_s \neq 0\}, \\ \sum_{s\in S} |\delta_s|, \\ \sum_{s\in S} \mathit{risk}_s \cdot \mathbf{1}\{\delta_s \neq 0\}, \\ -\sum_{s\in S} \mathit{benefit}_s \cdot \mathbf{1}\{\delta_s \neq 0\} \end{pmatrix} \tag{12}$$

Here, $\mathbf{1}\{\cdot\}$ denotes the indicator function, yielding 1 when the inner condition holds and 0 otherwise. The primary and secondary objectives minimize the number and total magnitude of altered clock bounds to produce concise repairs. The third term incorporates a penalty coefficient risk_s to discourage modifications prone to regression, while the final term uses a reward coefficient $\mathit{benefit}_s$ to prioritize repair sites that cover a broader set of violated traces.

C. Relation-Guided Search Optimization

During refinement, accumulating violated properties, TDTs, and editable bounds expands the MaxSMT search space. To reduce redundant constraints, MPTA-Repair uses relation-guided optimization to filter, group, and rank constraints and repair variables. These heuristics only guide the search; correctness is still guaranteed through full-property regression verification and an admissibility check.

At the property layer, timed safety properties are normalized into an `A[]` fragment over location predicates and clock or integer constraints. Equivalence is detected by the syntactic identity of normalized forms, while dominance is evaluated under logical subsumption, e.g., `A[]` $x \leq 5$ subsumes `A[]` $x \leq 10$

within the same structural context. MPTA-Repair then retains only the stronger properties for solving to optimize efficiency.

At the counterexample layer, MPTA-Repair maintains a TDT pool indexed by violated properties. Exact duplicates are removed, and the remaining traces are annotated with traversed transitions, visited locations, active clocks, and guard or invariant bounds. Traces with identical discrete transition sequences or overlapping clock zones are grouped to reveal common bottlenecks. A trace constraint is omitted from the active encoding only when it is an exact duplicate or soundly subsumed by an encoded trace constraint.

At the repair-bound layer, MPTA-Repair maps each active property-trace constraint to the clock bounds occurring in or affecting its encoding. This mapping restricts active MaxSMT decision variables to bounds relevant to the current counterexample pool and ranks them for candidate generation. Bounds with higher relational scores are prioritized because their modifications are more likely to satisfy the joint constraint system.

D. Validation, Admissibility, and Refinement

In MPTA-Repair, candidate validation determines whether a generated edit set can be accepted as a repair. After a candidate edit set is generated, the repaired model is verified against the entire property set Φ. If a candidate resolves the observed violation but induces new violations against the specification, it is rejected, indicating that the current property-trace pool is insufficient. MPTA-Repair then extracts the newly observed TDTs and appends them to the pool, guiding the subsequent refinement round.

Candidates that pass full-property regression verification are further evaluated against an admissibility check based on functional equivalence [8]. This step ensures that clock-bound modifications do not satisfy properties in ways that fail to preserve the original functional behavior. Rather than enforcing timed-language equivalence, which would conflict with the goal of eliminating violating timed paths, the check is performed by constructing untimed automata from both the original and repaired models and comparing their accepted untimed languages [16]. A candidate is rejected if it adds or removes discrete functional paths, ensuring that only timing characteristics are modified.

The refinement loop therefore combines local candidate generation with global validation. Local MaxSMT formulations propose candidate assignments, full-property regression verification ensures complete specification compliance, and the admissibility check preserves the original functional behavior.

IV. Experimental Evaluation

This section evaluates MPTA-Repair on both benchmark and industrial case studies. The evaluation follows the clock-bound repair setting defined in Section III, where only numerical bounds in guards and invariants are repairable. For comparison, we evaluate MPTA-Repair against an adapted iterative TARTAR-style baseline [8].

TABLE I
Statistics of the benchmark and industrial datasets.

Dataset	Original models	Mutated models	Avg. properties/model
Benchmark	9	54	4.1
Industrial	5	105	7.3

A. Evaluation Setup and Research Questions

We evaluate MPTA-Repair on two datasets, summarized in Table I. The benchmark dataset includes nine benchmark models derived from UPPAAL benchmark [14] and the XtaBenchmarkSuite [15], such as Elevator and FDDI. The industrial dataset constructed in this work contains five closely coupled NTA case-study models: Gear Control System for automotive transmission [17], SAE RoR for autonomous-driving decisions [18], Train Controller Alarm for railway emergency timing [19], Train Gate Controller for shared-resource concurrency [4], and Pacemaker for medical pacing [9]. Compared with the benchmark models, these case studies involve more tightly coupled timing constraints across multiple properties.

Since real-world NTA models with labeled clock-bound faults are difficult to obtain, we synthesize faulty instances using mutation [20], retaining only mutants that are syntactically valid, violate at least one timing safety property, and preserve the original functional behavior. Each repair instance is executed with a 10-minute timeout for both MPTA-Repair and the baseline.

We compare MPTA-Repair against an adapted iterative TARTAR-style baseline [8], which was originally designed for single-property repair and operates via trace-local repair generation. The evaluation is organized around the following research questions:

RQ1. How effective is MPTA-Repair compared with the iterative TARTAR-style baseline?

RQ2. How does repair effectiveness change from benchmark models to industrial NTA models with more tightly coupled timing constraints?

RQ3. What practical lessons do the results reveal about full-property validation, admissibility checking, and remaining repair failures?

B. Results by Research Question

RQ1: Overall repair effectiveness. Table II shows that MPTA-Repair outperforms the iterative baseline on both datasets. On the benchmark dataset, the baseline repairs 39 of 54 faulty instances, achieving a 72% success rate, whereas MPTA-Repair repairs 50 instances and reaches 93%. This result indicates that joint property-trace candidate generation improves repair effectiveness even when injected timing faults are relatively localized.

RQ2: Effect on industrial NTA models. The advantage of MPTA-Repair becomes more pronounced on the industrial dataset. The iterative baseline repairs only 21 of 105 faulty instances, corresponding to a 20% success rate, while MPTA-Repair repairs 80 instances and reaches 76%. The baseline reports a lower average duration on the industrial dataset

TABLE II
REPAIR RESULTS ACROSS BENCHMARK AND INDUSTRIAL DATASETS.

Dataset	Method	Mutated models	Repaired	Success rate	Avg. time
Benchmark	Iterative baseline	54	39	72%	60.61s
	MPTA-Repair	54	50	93%	15.46s
Industrial	Iterative baseline	105	21	20%	35.59s
	MPTA-Repair	105	80	76%	85.28s

because unsuccessful runs typically terminate early when no valid candidate can be found. Its success rate drops because trace-local repairs often produce candidates that fail full-property regression verification or the admissibility check under concurrent component interactions and tightly coupled timing constraints.

RQ3: Practical lessons and failure cases. MPTA-Repair achieves higher success rates by integrating constraints from TDTs and properties into a joint generation process. Consequently, the framework produces coordinated modifications that pass both full-property regression verification and the admissibility check. The results highlight two practical lessons for applying automated multi-property repair to NTA. First, full-property regression verification is necessary because industrial properties are often interdependent; a localized repair for one response-time property can inadvertently violate another safety property, such as maximum waiting bounds. MPTA-Repair avoids these regressions by requiring each candidate to satisfy the full property set. Second, the admissibility check is critical to distinguish valid repairs from behavior-breaking edits that satisfy timing properties only by removing intended functional behavior, such as preventing critical system actions.

Nevertheless, the framework can still fail to generate repairs in certain cases when it fails to find an admissible candidate within the time budget. This limitation is prominent in complex models like the Pacemaker case study, where tightly coupled timing constraints expand the search space, causing the solver to time out before finding a valid candidate.

V. CONCLUSION

This paper presented MPTA-Repair, an automated workflow for multi-property and multi-counterexample clock-bound repair of industrial NTA models. To overcome the limitations of isolated single-trace fixes, the method accumulates TDTs and exploits relations among properties, counterexamples, and repairable bounds to guide MaxSMT-based search. Each candidate is validated through full-property regression verification and an admissibility check based on functional equivalence [8], ensuring that accepted repairs preserve the original functional behavior.

Evaluations across benchmark and industrial datasets show that MPTA-Repair outperforms the iterative baseline, particularly in industrial systems with dense property interactions. The failure analysis further indicates that unsuccessful repairs mainly arise when the framework cannot find an admissible candidate within the time budget, especially as dense timing interactions enlarge the clock-bound repair search space.

Future work will primarily focus on extending the automated repair space beyond numerical clock bounds to include broader structural modifications, such as altering synchronization labels and redesigning discrete control logic. We also plan to integrate the admissibility check earlier in repair generation, so that functionally invalid candidates can be pruned before costly validation and large-scale industrial repair can be made more efficient.

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